

Figure 11.1. The one-loop diagram for the vacuum polarization in quantum electrodynamics. Here wavy lines represent photons; lines carrying arrows represent electrons.

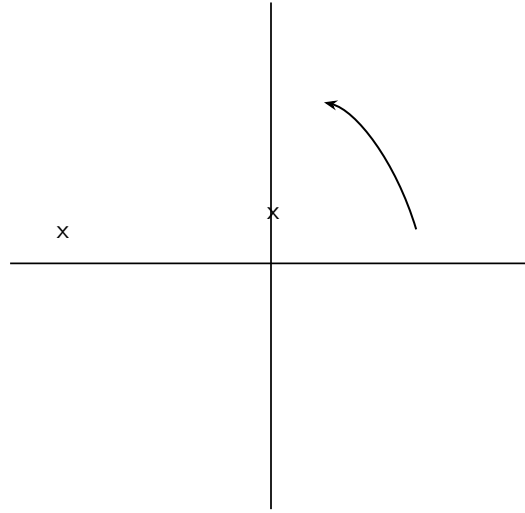


Figure 11.2. Wick rotation of the p^0 contour of integration. Small x's mark the poles in the p^0 complex plane; the arrow indicates the direction of rotation of the contour of integration, from the real to the imaginary p^0 -axes.

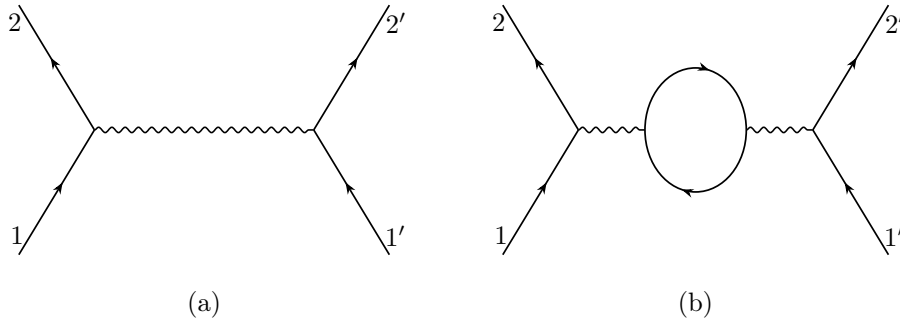


Figure 11.3. Two diagrams for the scattering of charged particles. Here lines carrying arrows are charged particles; wavy lines are photons. Diagram (b) represents the lowest-order vacuum polarization correction to the tree approximation graph (a).

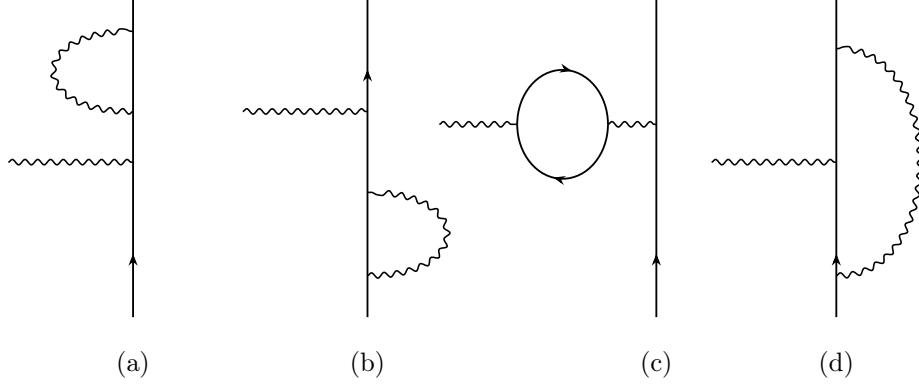


Figure 11.4. One-loop diagrams for the photon-lepton vertex function Γ^μ . Here wavy lines represent photons; other lines represent electrons or muons. Diagrams (a) and (b) are cancelled by lepton field renormalization terms; diagram (c) arises from the vacuum polarization calculated in Section 11.2; and (d) is the term calculated in Section 11.3.

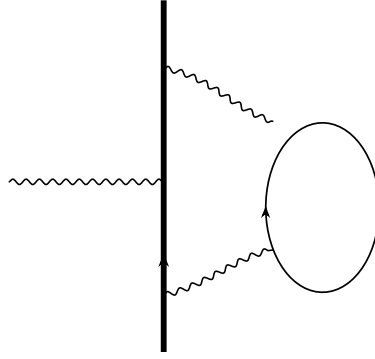


Figure 11.5. A two-loop diagram for the muon magnetic moment. Here the heavy straight line represents a muon; the light wavy lines are photons; and the other light lines are electrons. This diagram makes a relatively large contribution to the fourth-order muon gyromagnetic ratio, proportional to $\ln(m_\mu/m_e)$.

1 One-Loop Radiative Corrections in Quantum Electrodynamics

1.1 Electron Self-Energy

We conclude this chapter with a calculation of the electron self-energy function. This by itself does not have any direct experimental implications, but some of the results here will be useful in Chapter 14 and Volume II.

As in Section 10.3, we define $i(2\pi)^2[\Sigma^*(p)]_{\beta,\alpha}$ as the sum of all graphs with one incoming and one outgoing electron line carrying momenta p and Dirac indices α and β respectively, with the asterisk indicating that we exclude diagrams that can be disconnected by cutting through some internal electron line, and with propagators omitted for the two external

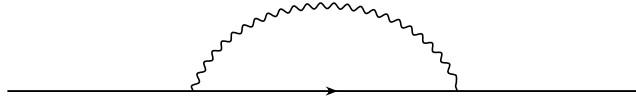


Figure 11.6. The one-loop diagram for the electron self-energy function. As usual, the straight line represents an electron; the wavy line is a photon.

lines. The complete electron propagator is then given by the sum

$$[-i(2\pi)^{-4}S'(p)] = [-i(2\pi)^{-4}S(p)] + [-i(2\pi)^{-4}S(p)][i(2\pi)^4\Sigma^*(p)][-i(2\pi)^{-4}S(p)] + \cdots, \quad (11.4.1)$$

where

$$S(p) = \frac{-i \not{p} + m_e}{p^2 + m_e^2 - i\epsilon}. \quad (11.4.2)$$

The sum is trivial, and gives

$$S'(p) = [i \not{p} + m_e - \Sigma^*(p) - i\epsilon]^{-1}. \quad (11.4.3)$$

In lowest order there is a one-loop contribution to Σ^* , given by Figure 11.6:

$$\begin{aligned} i(2\pi)^4\Sigma_{1\text{ loop}}^*(p) &= \int d^4k \left[\frac{-i}{(2\pi)^4} \frac{\eta_{\rho\sigma}}{k^2 - i\epsilon} \right] \times [(2\pi)^4 e\gamma^\rho] \\ &\times \left[\frac{-i}{(2\pi)^4} \frac{-i \not{p} + i \not{k} + m_e}{(p-k)^2 + m_e^2 - i\epsilon} \right] [(2\pi)^4 e\gamma^\sigma] \end{aligned}$$

or more simply

$$\Sigma_{1\text{ loop}}^*(p) = \frac{ie^2}{(2\pi)^4} \int d^4k \left[\frac{1}{k^2 - i\epsilon} \right] \left[\frac{\gamma^\rho (-i \not{p} + i \not{k} + m_e) \gamma_\rho}{(p-k)^2 + m_e^2 - i\epsilon} \right]. \quad (11.4.4)$$

(This is in Feynman gauge; amplitudes with charged particles off the mass shell are not gauge-invariant.) For use in our calculation of the Lamb shift, it will be convenient to use a method of regularization introduced by Pauli and Villars [8]. We replace the photon propagator $(k^2 - i\epsilon)^{-1}$ with

$$\frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon},$$

so that the electron self-energy function is

$$\begin{aligned} \Sigma_{1\text{ loop}}^*(p) &= \frac{ie^2}{(2\pi)^4} \int d^4k \left[\frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon} \right] \\ &\times \left[\frac{\gamma^\rho (-i \not{p} + i \not{k} + m_e) \gamma_\rho}{(p-k)^2 + m_e^2 - i\epsilon} \right]. \end{aligned} \quad (11.4.5)$$

Later we can drop the regulator by letting the regulator mass μ go to infinity. In Chapter 14 we will also be interested in the case where $\mu \ll m_e$.

We again use the Feynman trick to combine denominators, and recall that $\gamma^\rho \gamma^\kappa \gamma_\rho = -2\gamma^\kappa$ and $\gamma^\rho \gamma_\rho = 4$. This gives

$$\begin{aligned} \Sigma_{1 \text{ loop}}^*(p) &= \frac{ie^2}{(2\pi)^4} \int d^4k [2i(\not{p} - \not{k}) + 4m_e] \\ &\quad \times \int_0^1 dx \left[\frac{1}{\{(k - px)^2 + p^2x(1-x) + m_e^2x - i\epsilon\}^2} \right. \\ &\quad \left. - \frac{1}{\{(k - px)^2 + p^2x(1-x) + m_e^2x + \mu^2(1-x) - i\epsilon\}^2} \right]. \end{aligned} \quad (11.4.6)$$

Shifting the variable of integration $k \rightarrow k + px$ and rotating the contour of integration gives

$$\begin{aligned} \Sigma_{1 \text{ loop}}^*(p) &= \frac{-2\pi^2 e^2}{(2\pi)^4} \int_0^1 dx [2i(1-x) \not{p} + 4m_e] \int_0^\infty d\kappa \kappa^3 \\ &\quad \times \left[\frac{1}{\{\kappa^2 + p^2x(1-x) + m_e^2x\}^2} - \frac{1}{\{\kappa^2 + p^2x(1-x) + m_e^2x + \mu^2(1-x)\}^2} \right]. \end{aligned} \quad (11.4.7)$$

The κ -integral is trivial:

$$\Sigma_{1 \text{ loop}}^*(p) = \frac{-\pi^2 e^2}{(2\pi)^4} \int_0^1 dx [2i(1-x) \not{p} + 4m_e] \ln \left(\frac{p^2x(1-x) + m_e^2x + \mu^2(1-x)}{p^2x(1-x) + m_e^2x} \right). \quad (11.4.8)$$

The interaction (11.1.9) also contributes a renormalization counterterm $-(Z_2 - 1)(i \not{p} + m_e) + Z_2 \delta m_e$ in $\Sigma^*(p)$, with Z_2 and δm_e determined by the condition that the complete propagator $S'(p)$ regarded as a function of $i \not{p}$ should have a pole at $i \not{p} = -m_e$ with residue unity. (As we shall see in the next chapter, this makes Σ^* finite as $\mu \rightarrow \infty$ to all orders in e .) In lowest order, this gives

$$\begin{aligned} \delta m_e &= - \Sigma_{1 \text{ loop}}^* \Big|_{i\not{p} = -m_e} \\ &= \frac{2m_e \pi^2 e^2}{(2\pi)^4} \int_0^1 dx [1+x] \ln \left(\frac{m_e^2 x^2 + \mu^2(1-x)}{m_e^2 x^2} \right), \end{aligned} \quad (11.4.9)$$

$$\begin{aligned} Z_2 - 1 &= -i \frac{\partial \Sigma_{1 \text{ loop}}^*}{\partial \not{p}} \Big|_{i\not{p} = -m_e} \\ &= \frac{-2\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \left\{ (1-x) \ln \left(\frac{m_e^2 x^2 + \mu^2(1-x)}{m_e^2 x^2} \right) - \frac{2\mu^2(1-x)^2(1+x)}{x[m_e^2 x^2 + \mu^2(1-x)]} \right\}. \end{aligned} \quad (11.4.10)$$

(To this order, we do not distinguish between δm_e and $Z_2 \delta m_e$.) Dropping terms that vanish for $\mu^2 \rightarrow \infty$, Eqs. (11.4.8)–(11.4.10) yield

$$\Sigma_{1 \text{ loop}}^*(p) = \frac{-\pi^2 e^2}{(2\pi)^4} \int_0^1 dx [2i(1-x) \not{p} + 4m_e] \ln \left(\frac{\mu^2(1-x)}{p^2x(1-x) + m_e^2x} \right), \quad (11.4.11)$$

$$\delta m_e = \frac{2m_e\pi^2 e^2}{(2\pi)^4} \int_0^1 dx [1+x] \ln \left(\frac{\mu^2(1-x)}{m_e^2 x^2} \right), \quad (11.4.12)$$

$$Z_2 - 1 = \frac{-2\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \left\{ (1-x) \ln \left(\frac{\mu^2(1-x)}{m_e^2 x^2} \right) - \frac{2(1-x^2)}{x} \right\}. \quad (11.4.13)$$

Inspection then shows that in the complete self-energy function the $\ln \mu^2$ terms cancel, leaving us with

$$\begin{aligned} \Sigma_{\text{order } e^2}^*(p) &= \Sigma_{1 \text{ loop}}^*(p) - (Z_2 - 1)(i \not{p} + m_e) + Z_2 \delta m_e \\ &= \frac{2\pi^2 e^2}{(2\pi)^4} \int_0^1 dx \left\{ [i(1-x) \not{p} + 2m_e] \ln \left(\frac{m_e^2(1-x)}{p^2 x(1-x) + m_e^2 x} \right) \right. \\ &\quad \left. - m_e[1+x] \ln \left(\frac{1-x}{x^2} \right) \right. \\ &\quad \left. - (i \not{p} + m_e) \left[(1-x) \ln \left(\frac{1-x}{x^2} \right) - \frac{2(1-x^2)}{x} \right] \right\}. \end{aligned} \quad (11.4.14)$$

There is still a divergence from the behavior of the last term as $x \rightarrow 0$, which can be traced to the singular behavior of the integral over the photon momentum k in Eq. (11.4.5) at $k^2 = 0$, when we take p^2 at the point $p^2 = -m_e^2$ where we evaluated $Z_2 - 1$. Such infrared divergences will be discussed in detail in Chapter 13. For the present, the point that concerns us is that the ultraviolet divergence has cancelled.

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The result (11.4.9) for δm_e is of some interest in itself. Note that $\delta m_e/m_e > 0$, as we would expect for the electromagnetic self-energy due to the interaction of a charge with its own field. But unlike the classical estimates of electromagnetic self-energy by Poincaré, Abraham, and others [9], Eq. (11.4.9) is only logarithmically divergent in the limit $\mu \rightarrow \infty$ where the cutoff is removed. In this limit:

$$\delta m_e \rightarrow \frac{6m_e\pi^2 e^2}{(2\pi)^4} \ln \left(\frac{\mu}{m_e} \right). \quad (11.4.15)$$

In our calculation of the Lamb shift in Section 14.3 we will be interested in the opposite limit, $\mu \ll m_e$. Here Eq. (11.4.9) gives

$$\delta m_e \rightarrow \frac{e^2 \mu}{8\pi} \left[1 - \frac{\mu}{2\pi m_e} + \cdots \right]. \quad (11.4.16)$$

Appendix Assorted Integrals

In order to combine the denominators of N propagators, we need to replace a product like $D_1^{-1} D_2^{-1} \cdots D_N^{-1}$ with an integral of a function that involves a linear combination of $D_1, D_2, \dots, D_N$. For this purpose it is often convenient to make use of the formula

$$\begin{aligned} \frac{1}{D_1 D_2 \cdots D_N} &= (N-1)! \int_0^1 dx_1 \int_0^{x_1} dx_2 \cdots \int_0^{x_{N-2}} dx_{N-1} \\ &\quad \times [D_1 x_{N-1} + D_2 (x_{N-2} - x_{N-1}) + \cdots + D_N (1 - x_1)]^{-N}. \end{aligned} \quad (11.A.1)$$

In this chapter we have used special cases of this formula for $N = 2$ and $N = 3$.

After combining denominators, shifting the four-momentum variable of integration, Wick rotating, and using four-dimensional rotational invariance, we commonly encounter integrals of the form

$$\int d^4k \frac{(k^2)^n}{(k^2 + \nu^2)^m}$$

with $(k^2 + \nu^2)^m$ coming from the combined propagator denominators, and $(k^2)^n$ coming from the propagator numerators and vertex momentum factors. This is divergent for $2n + 4 \geq 2m$, but the integral can be given a finite value by analytically continuing the spacetime dimensionality from 4 to a complex value d . To evaluate the resulting integral, we use the well-known formula

$$\int_0^\infty d\kappa \frac{\kappa^{\ell-1}}{(\kappa^2 + \nu^2)^m} = \nu^{\ell-2m} \frac{\Gamma(\ell/2)\Gamma(m-\ell/2)}{2\Gamma(m)}, \quad (11.A.2)$$

where $\ell = d + 2n$. We used this formula in the special cases $n = 0, m = 2$ and $n = 1, m = 2$ in Section 11.2.

Ultraviolet divergences manifest themselves in Eq. (11.A.2) as poles in the factor $\Gamma(m - \ell/2) = \Gamma(m - n - d/2)$ as $d \rightarrow 4$ with fixed integer n . For $2 + n = m$, this factor goes as

$$\Gamma\left(\frac{4-d}{2}\right) \rightarrow \frac{2}{d-4} + \gamma, \quad (11.A.3)$$

where $\gamma = 0.5772157 \dots$ is the Euler constant. The limiting behavior for $2 + n > m$ can be obtained from (11.A.3) and the recursion relation for Gamma functions.